

SIMATS ENGINEERING ASSESSMENT TOOLS

COURSE NAME: Theory of Computation

COURSE CODE: CSA13

CO5-AT①

COURSE OUTCOME COVERED

CO5: Design Pushdown Automata and analyze their equivalence with Context-Free Grammars through formal construction and proof techniques. (BL :3)

Assessment Tool Weightage: 40%

QUESTIONS: 5
Duration : 1 Hour

Total Marks:25

Constraint Based Problem

Code of Conduct:

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I Omy Sai Reddy T (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



SIMATS ENGINEERING ASSESSMENT TOOLS

1. Design a Pushdown Automaton that accepts the language

$$L = \{ a^n b^n \mid n \geq 1 \}$$

Constraint: You are allowed to use only one stack symbol (besides initial symbol)

2. CFG to PDA Conversion with Limited Productions

Given a Context-Free Grammar:

$$S \rightarrow aSb \mid ab$$

Construct an equivalent PDA.

Constraint: PDA must be constructed using only 2 states

3. PDA to CFG Conversion with Restricted Variables

Convert a given PDA into an equivalent CFG.

Constraint: The resulting CFG must use at most 3 variables (non-terminals)

4. Turing Machine with Limited Tape Movement

Design a Turing Machine to recognize the language

$$L = \{ ww \mid w \in \{0,1\}^* \}$$

Constraint: The tape head can move only to the right (no left movement allowed)

5. Consider a language $L = \{ a^n b^n c^n \mid n \geq 1 \}$

Constraint: Design a Turing Machine for L

for $L = \{a^n b^n \mid n \geq 1\}$

constraint:-

$\neq \neq \neq$

Any one select stack symbol decides the initial stack symbol is allowed:-

Let the stack symbol

- A = the only additional stack symbol.
- Z_0 = initial stack symbol.

Idea:-

- * For every a, Push one A onto the stack.
- * For every b, Pop one A.
- * Accept when all A's been removed and only Z_0 remains.

$$M = [Q, \Sigma, \Gamma, \delta, q_0, Z_0, F]$$

Where $Q = \{q_0, q_1, q_2\}$

$\Sigma = \{a, b\}$

$\Gamma = \{Z_0, A\}$

$F = \{q_f\}$

Transition Function.

CS	Input	Stack Top	Operation	New state
q_0	a	Z_0	Push A	q_0
q_0	a	A	Push A	q_0
q_0	b	A	Pop A	q_1
q_1	b	A	Pop A	q_1
q_1	ϵ	Z_0	Accept	q_f

Formal Funs:-

$$\delta(q_0, a, z_0) = (q_0, A z_0)$$

$$\delta(q_0, a, A) = (q_0, AA)$$

$$\delta(q_0, b, A) = (q_1, \epsilon)$$

$$\delta(q_1, b, A) = (q_1, \epsilon)$$

$$\delta(q_1, \epsilon, z_0) = (q_f, z_0).$$

Conclusion:-

The PDA pushes one A for every a and pops one A for every b therefore it accepts exactly.

$$L = \{a^n b^n \mid n \geq 1\}$$

Using only one additional stack symbol A.

CFG to PDA Conversion:-

Given CFG

$$S \rightarrow a s b \mid a b$$

Constraint:- The PDA must use only 2 states.

PDA Construction:-

let:-

$$Q = \{q_0, q_1\}$$

• q_0 : Processing state

• q_1 : Accepting state,

Stack alphabet: $M = \{z_0, s, a, b\}$

Input alphabet: $\Sigma = \{a, b\}$

Step ①:- Push the start value: $\delta(q_0, \epsilon, z_0) = (q_0, s, z_0)$

Step ②:- Replace using grammar productions.

For:- $S \rightarrow asb$

$$\delta(q_0, \epsilon, \$) = (q_0, b\$a)$$

$$S \rightarrow ab$$

$$\delta(q_0, q_s) = (q_0, ba)$$

Step ③:- Match terminals:- $\delta(q_0, a, a) = (q_0, \epsilon)$

$$\delta(q_0, b, b) = (q_0, \epsilon)$$

Step ④:- Accept:- $\delta(q_0, \epsilon, z_0) = (q_1, z_0)$

Conclusions:- The PDA uses only 2 states and simulate productions

$$S \rightarrow asb | ab$$

Thus, PDA accepts string such as: $ab, aabb, aaabbb, \dots$

③ PDA to CFG Conversion with at most 3 Variables:-

General Approach:-

Suppose PDA has states q_0 and q_1

- * Reading a's and pushing a stack symbol.
- * Reading b's and popping the stack symbol.
- * Accepting when stack becomes empty.

$$L = \{a^n b^n \mid n \geq 1\}$$

$$CFG = S \rightarrow asb | ab$$

$$\text{Variables } V = \{S\}$$

only one terminal is actually required, which satisfies

The constraint of at most 3 Variables.

Explanation:- $S \rightarrow asb$
 $S \rightarrow ab$

$$S \rightarrow asb$$

$$S \rightarrow aasbb$$

$$S \rightarrow aaabbb$$

Thus Equivalent CFG is

$$\therefore S \rightarrow asb/ab.$$

which is uses only one terminal.

④ **Pushing Machine For** $L = \{ww \mid w \in \{0,1\}^*\}$

Constraint :- The tape head can move only to the right.

Such a turning Machine cannot recognise $L = \{ww\}$.

Reason :- A standard one-tape turning machine that can move only to the right cannot move back to.

* Compare symbols in the first half with corresponding symbols in second half.

* Mark previously checked symbols.

* Return to an earlier portion.

For Ex:-
= * =

$$w = 01$$

$$ww = 0101.$$

* First 0 with third 0.

* Second 1 with fourth 1.

⇒ This requires remembering an arbitrary of information about the first half (or) moving back on tape.

⇒ A Machine restricted only right movement behave like finite automation, because, once it's moves, apart a symbol. It cannot inspect it again.

$$L = \{ww \mid w \in \{0,1\}^*\}$$

is not a regular language.

So, standard one tape right only turning Machine can recognize $\{ww\}$.

Final Answer:-

The given constraint makes the problem impossible. A turning machine must be allowed to move both left and right.

⑤ **Turning Machine for $L = \{a^n b^n c^n \mid n \geq 1\}$**

Idea:-

$= * =$

The turning Machine repeatedly:-

- * Find the leftmost unmarked a, marks it as x.
- * Moves right and find the first unmarked b, marking it as y.
- * Moves right and finds first unmarked c, marking it as z.
- * Returns to beginning.
- * Repeat until all symbols are marked.
- * Accepts it all a, b and c symbols are marked.

Tape Symbols:-

$$M = \{a, b, c, x, y, z, B\}$$

where:-

- * x marks a processed 'a'.
- * y marks a processed 'b'.
- * z marks a processed 'c'.
- * B (marks a processed) blank.

working Procedure :-

$a \rightarrow x$

$b \rightarrow y$

$c \rightarrow z$

Ex:-

input = a a a b b b c c c
x a a b b b c c c
x a a x b b c c c
x a a a y b b z c c.

Eventually x x x y y y z z z.

Conclusion:-

The turning Machine Accepts:-

$$L = \{ a^n b^n c^n \mid n \geq 1 \}$$

SIMATS ENGINEERING ASSESSMENT TOOLS

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understandin g
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well- organized answer	Understandabl e with minor issues	Somewhat unclear or poorly structured	Difficult to understand

92/100

JH